

## 1. General Description

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The EMS3515 is a high-performance single-pole single-throw (SPST) depletion switch. Depletion technology enables the device to conduct signals in the absence of  $V_{DD}$  and to isolate signals in the presence of  $V_{DD}$ . The EMS3515 can pass or isolate negative signal fluctuation as low as -4.2 V and is 5.5 V tolerable. The EMS3515 's depletion gate control enables it to achieve exceptional THD+N performance while using very little power during signal conduction.

## 2. Features and Benefits

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- Normally closed when  $V_{DD} < 0.5$  V
- $V_{SW}$ : - 4.2 V to 5.5 V
- $R_{ON}$ : 0.85  $\Omega$  (typ.)
- $R_{FLAT(ON)}$ : 8.7 m $\Omega$  (typ.)
- THD+N: - 102 dB (typ.)

## 3. Applications

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- Smart phones, Tablets, laptops
- Bluetooth headsets, Mobile accessories

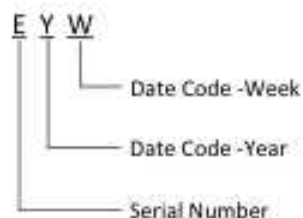
## 4. Ordering Information

Table 1. Ordering information

| Type number | Topside marking | Package |                                                                                   |          |
|-------------|-----------------|---------|-----------------------------------------------------------------------------------|----------|
|             |                 | Name    | Description                                                                       | Quantity |
| EMS3515YFQ  | E<br>YW         | WLCSP4  | WLCSP package, 4 bumps, 0.93 mm × 0.93 mm;<br>0.4 mm pitch, 0.445 mm (Typ) height | 3000     |

### TOPSIDE MARKING INFORMATION

NOTE: YW = Date Code.



## 5. Function Diagram

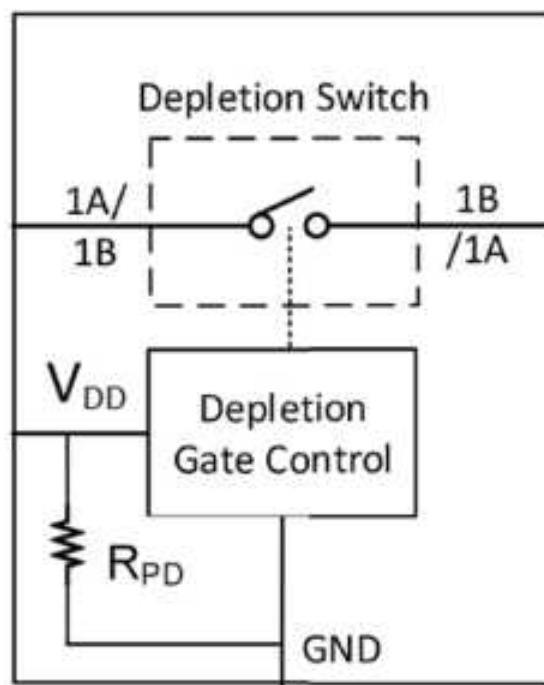


Fig 1. Function Diagram

## 6. Pinning Information

### 6.1. Pin map

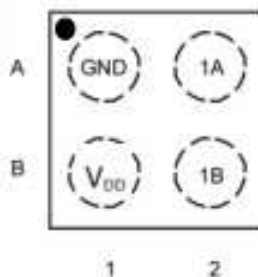


Fig 2. Top view pin configuration

### 6.2. Pin description

Table 2. Pin description

| Symbol          | Pin | Description                                                      |
|-----------------|-----|------------------------------------------------------------------|
| GND             | A1  | Ground                                                           |
| 1A              | A2  | A-port of switch                                                 |
| V <sub>DD</sub> | B1  | Supply voltage (switch is conducted when V <sub>DD</sub> is low) |
| 1B              | B2  | B-port of switch                                                 |

## 7. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

| V <sub>DD</sub> | Switch state    |
|-----------------|-----------------|
| L               | ON (Conducting) |
| H               | OFF (Isolating) |

## 8. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

**Table 4. Absolute Maximum Ratings**

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

| Symbol    | Parameter                                                                    | Conditions                                |            | Min     | Max | Unit |
|-----------|------------------------------------------------------------------------------|-------------------------------------------|------------|---------|-----|------|
| $V_{DD}$  | Supply voltage                                                               |                                           |            | - 0.5   | 6   | V    |
| $V_{SW}$  | Switch voltage range                                                         | DC switch I/O voltage (switch conducting) |            | - 4.5   | 6   | V    |
|           |                                                                              | DC switch I/O voltage (switch isolated)   |            | - 4.5   | 6   | V    |
| $I_{SW}$  | Maximum peak switch I/O current<br>- Pulsed at 1ms duration, <10% duty cycle |                                           |            |         | 500 | mA   |
| $T_{stg}$ | Storage Temperature                                                          |                                           |            | - 65    | 150 | °C   |
| ESD       | ESD Capability                                                               | HBM, JEDEC :<br>JESD22-A114               | I/O to GND | $\pm 6$ |     | kV   |
|           |                                                                              |                                           | Others     | $\pm 7$ |     | kV   |

## 9. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

**Table 5. Recommended Operating Conditions**

| Symbol        | Parameter           | Conditions                                                     | Min  | Max | Unit             |
|---------------|---------------------|----------------------------------------------------------------|------|-----|------------------|
| $V_{DD}$      | Supply voltage      | Isolating                                                      | 2.0  | 5.5 | V                |
| $V_{SW(OFF)}$ | Switch voltage      | Isolating ( $V_{DD}=2.0V$ , $R_L=32\Omega$ , $f<30kHz$ )       |      | 1.5 | V <sub>rms</sub> |
|               |                     | Isolating ( $V_{DD}=2.5V$ , $R_L=32\Omega$ , $f<30kHz$ )       |      | 1.8 | V <sub>rms</sub> |
|               |                     | Isolating ( $V_{DD}=3.0V$ , $R_L=32\Omega$ , $f<30kHz$ )       |      | 2.4 | V <sub>rms</sub> |
|               |                     | Isolating ( $V_{DD}=3.5V$ , $R_L=32\Omega$ , $f<30kHz$ )       |      | 2.6 | V <sub>rms</sub> |
|               |                     | Isolating ( $V_{DD}=4.0V$ , $R_L=32\Omega$ , $f<30kHz$ )       |      | 2.7 | V <sub>rms</sub> |
|               |                     | Isolating ( $V_{DD}=4.5V$ , $R_L=32\Omega$ , $f<30kHz$ )       |      | 2.8 | V <sub>rms</sub> |
|               |                     | Isolating ( $V_{DD}=5.0V$ , $R_L=32\Omega$ , $f<30kHz$ )       |      | 3.0 | V <sub>rms</sub> |
|               |                     | Isolating (requires 1 k $\Omega$ (typ.) in series with source) | 4.7  | 5.5 | V                |
| $V_{SW(ON)}$  |                     | Conducting ( $f<30kHz$ )                                       |      | 3.0 | V <sub>rms</sub> |
|               |                     | Isolating (requires 1 k $\Omega$ (typ.) in series with source) | 4.7  | 5.5 | V                |
| $T_A$         | Ambient Temperature |                                                                | - 40 | 85  | °C               |

## 10. Electrical Characteristics

**Table 6. Static characteristics**

Unless otherwise specified, typical values are at  $V_{DD} = 0\text{ V}$ ,  $T_A = +25\text{ }^{\circ}\text{C}$ .

| Symbol                         | Parameter                            | Conditions                                                                                    | Min                   | Typ   | Max | Unit |
|--------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------|-------|-----|------|
| DC Electrical Characteristics  |                                      |                                                                                               |                       |       |     |      |
| I <sub>OFF</sub>               | Switch off leakage current           | V <sub>DD</sub> = 3.3 V to 4.5 V,<br>1B = GND                                                 | 1A = V <sub>DD</sub>  | 0.2   |     | μA   |
|                                |                                      |                                                                                               | 1A = - 1.5 V to 1.5 V | 0.7   |     |      |
|                                |                                      | V <sub>DD</sub> = 2 V to 3.3 V,<br>1B = GND                                                   | 1A = - 1.5 V to 1.5 V | 0.5   |     | μA   |
| R <sub>ON</sub>                | Switch on resistance                 | I <sub>SW</sub> = 50 mA, V <sub>SW</sub> = - 3 V to 3 V                                       |                       | 0.85  |     | Ω    |
| R <sub>FLAT(ON)</sub>          | On resistance flatness               | I <sub>SW</sub> = 50 mA, V <sub>SW</sub> = - 3 V to 3 V                                       |                       | 8.7   |     | mΩ   |
| C <sub>ON</sub>                | On capacitance                       | V <sub>SW</sub> = 0 dBm, f = 1 MHz                                                            |                       | 17    |     | pF   |
| C <sub>OFF</sub>               | Off capacitance                      | V <sub>SW</sub> = 0 dBm, f = 1 MHz                                                            |                       | 25    |     | pF   |
| AC Electronics Characteristics |                                      |                                                                                               |                       |       |     |      |
| THD+N                          | Total harmonic distortion plus noise | V <sub>SW</sub> = 0.7 V <sub>RMS</sub> , Ground centered<br>R <sub>L</sub> = 32 Ω , f = 1 kHz |                       | - 102 |     | dB   |
| OIRR                           | Off isolation rejection ratio        | V <sub>DD</sub> = 2.0V<br>V <sub>sw</sub> < 1.7V <sub>rms</sub> , f = 20kHz                   | R <sub>L</sub> = 32Ω  | -115  |     | dB   |
|                                |                                      |                                                                                               | R <sub>L</sub> = 1kΩ  | -100  |     | dB   |
|                                |                                      | V <sub>DD</sub> = 2.5V<br>V <sub>sw</sub> < 2.0V <sub>rms</sub> , f = 20kHz                   | R <sub>L</sub> = 32Ω  | -115  |     | dB   |
|                                |                                      |                                                                                               | R <sub>L</sub> = 1kΩ  | -100  |     | dB   |
|                                |                                      | V <sub>DD</sub> = 3.0V<br>V <sub>sw</sub> <2.4V <sub>rms</sub> , f = 20kHz                    | R <sub>L</sub> = 32Ω  | -115  |     | dB   |
|                                |                                      |                                                                                               | R <sub>L</sub> = 1kΩ  | -100  |     | dB   |
|                                |                                      | V <sub>DD</sub> = 3.5V<br>V <sub>sw</sub> <2.6V <sub>rms</sub> , f = 20kHz                    | R <sub>L</sub> = 32Ω  | -115  |     | dB   |
|                                |                                      |                                                                                               | R <sub>L</sub> = 1kΩ  | -100  |     | dB   |
|                                |                                      | V <sub>DD</sub> = 4.0V<br>V <sub>sw</sub> < 2.7V <sub>rms</sub> , f = 20kHz                   | R <sub>L</sub> = 32Ω  | -115  |     | dB   |
|                                |                                      |                                                                                               | R <sub>L</sub> = 1kΩ  | -108  |     | dB   |
|                                |                                      | V <sub>DD</sub> = 4.5V<br>V <sub>sw</sub> < 2.8V <sub>rms</sub> , f = 20kHz                   | R <sub>L</sub> = 32Ω  | -115  |     | dB   |
|                                |                                      |                                                                                               | R <sub>L</sub> = 1kΩ  | -108  |     | dB   |
|                                |                                      | V <sub>DD</sub> = 5.0V<br>V <sub>sw</sub> < 3.0V <sub>rms</sub> , f = 20kHz                   | R <sub>L</sub> = 32Ω  | -115  |     | dB   |
|                                |                                      |                                                                                               | R <sub>L</sub> = 1kΩ  | -108  |     | dB   |

## EMS3515

### 5 V SPST Depletion Switch with Negative Swing

| Symbol                           | Parameter                     | Conditions                                                                                                                        | Min | Typ | Max | Unit          |
|----------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|---------------|
| BW                               | Bandwidth                     | $R_L = 50 \Omega$ , $C_L = 0 \text{ pF}$                                                                                          |     | 80  |     | MHz           |
| <b>Supply currents</b>           |                               |                                                                                                                                   |     |     |     |               |
| $I_{DD}$                         | Quiescent current             | $V_{DD} = 2.0 \text{ V}$                                                                                                          |     | 30  |     | $\mu\text{A}$ |
|                                  |                               | $V_{DD} = 2.5 \text{ V}$                                                                                                          |     | 37  |     | $\mu\text{A}$ |
|                                  |                               | $V_{DD} = 3.3 \text{ V}$                                                                                                          |     | 46  |     | $\mu\text{A}$ |
|                                  |                               | $V_{DD} = 4.2 \text{ V}$                                                                                                          |     | 62  |     | $\mu\text{A}$ |
|                                  |                               | $V_{DD} = 5.0 \text{ V}$                                                                                                          |     | 79  |     | $\mu\text{A}$ |
| $I_{DIS}$                        | Disable current               | $V_{DD} \leq 0.2 \text{ V}$                                                                                                       |     | 0.1 |     | $\mu\text{A}$ |
| <b>Control logic</b>             |                               |                                                                                                                                   |     |     |     |               |
| $R_{PD}$                         | $V_{DD}$ pull-down resistance | $V_{DD} \leq 0.2 \text{ V}$                                                                                                       |     | 5.5 |     | M $\Omega$    |
| $V_{DDH}$                        | $V_{DD}$ high voltage         |                                                                                                                                   | 2.0 |     |     | V             |
| $V_{DDL}$                        | $V_{DD}$ low voltage          |                                                                                                                                   |     |     | 0.4 | V             |
| <b>Switching characteristics</b> |                               |                                                                                                                                   |     |     |     |               |
| $t_{OFF}$                        | Switch turn-off Time          | $R_L = 1 \text{ k}\Omega$ , $C_L = 10 \text{ pF}$ ,<br>$V_{DD} = 0.0 \text{ V to } 2.0 \text{ V}$ ,<br>$V_{SW} = 1.5 \text{ V}$ ; |     | 170 |     | $\mu\text{s}$ |
| $t_{ON}$                         | Switch turn-on Time           | $R_L = 1 \text{ k}\Omega$ , $C_L = 10 \text{ pF}$ ,<br>$V_{DD} = 2.0 \text{ V to } 0 \text{ V}$ ,<br>$V_{SW} = 1.5 \text{ V}$ ;   |     | 1   |     | $\mu\text{s}$ |
| <b>Oscillator frequency</b>      |                               |                                                                                                                                   |     |     |     |               |
| $f_{OSC}$                        | On-chip oscillator frequency  | For reference only                                                                                                                |     | 3.3 |     | MHz           |

## 11. Typical Characteristics

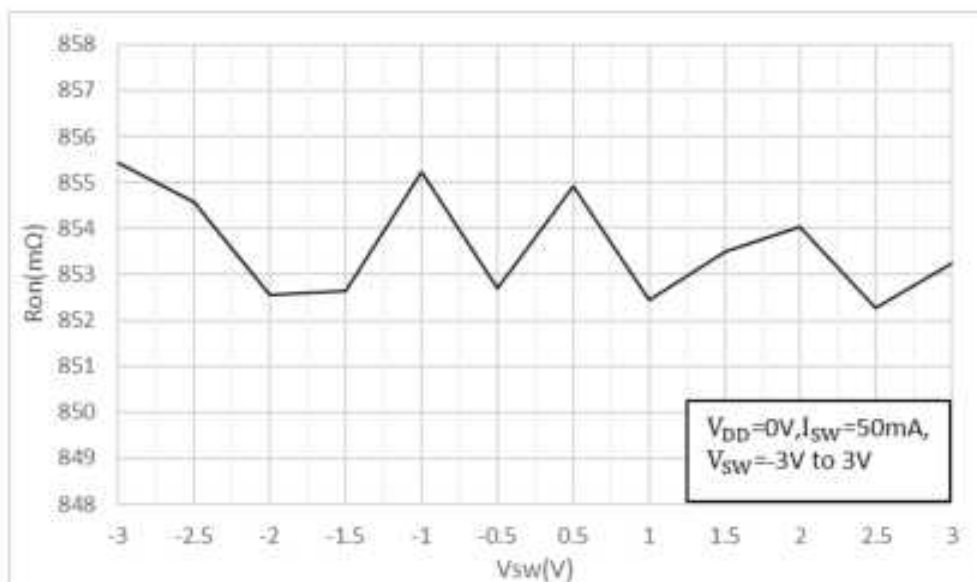


Fig 3.  $R_{ON}$  vs.  $V_{SW}$

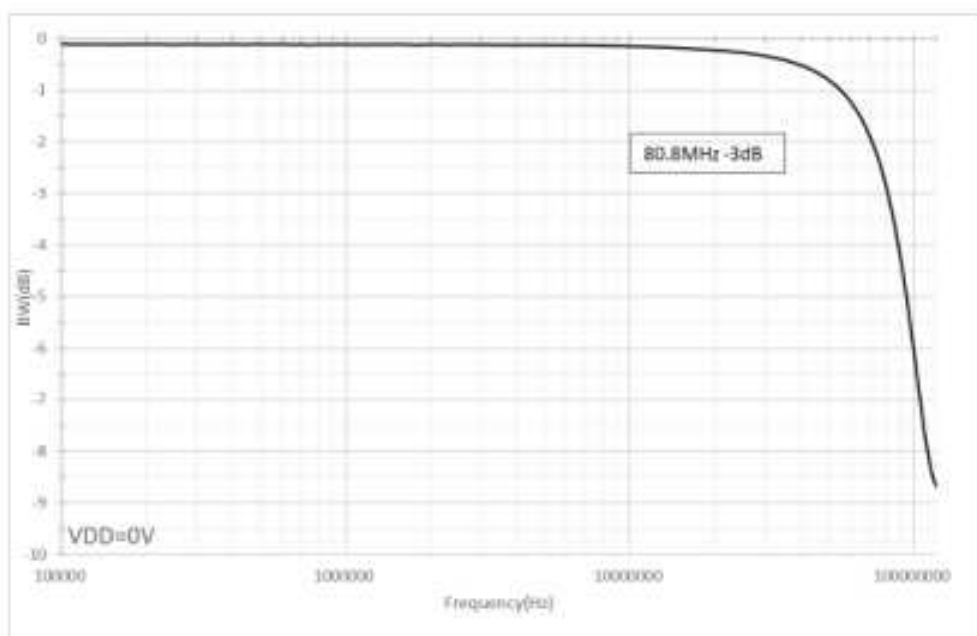


Fig 4. BW vs. Frequency

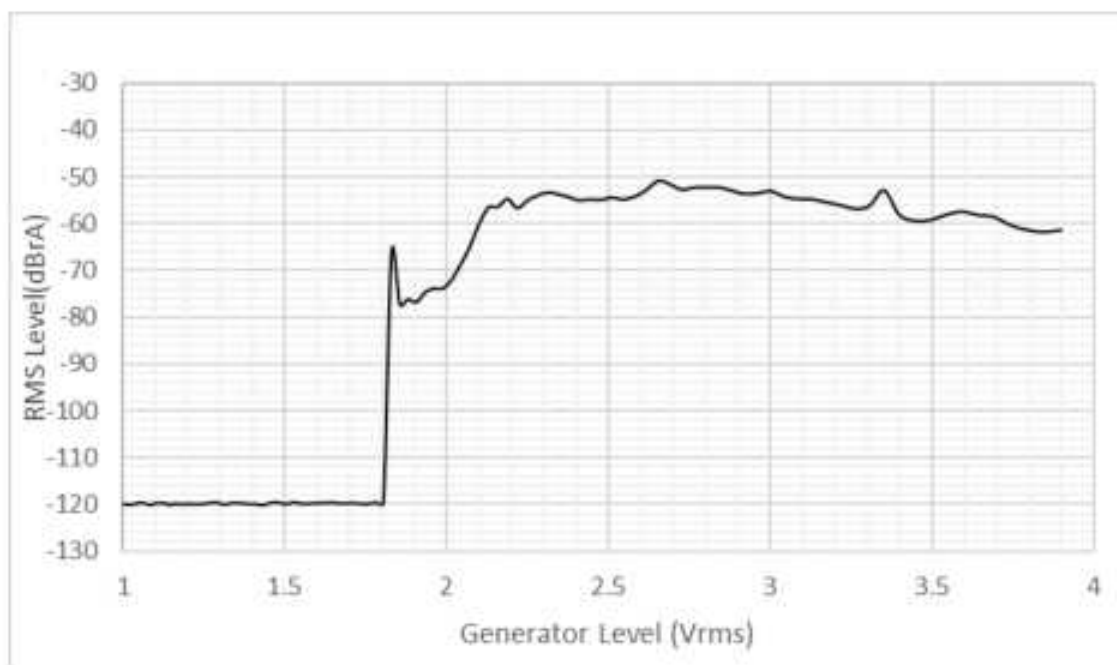


Fig 5. THD + N vs. Input with 33Ω Load

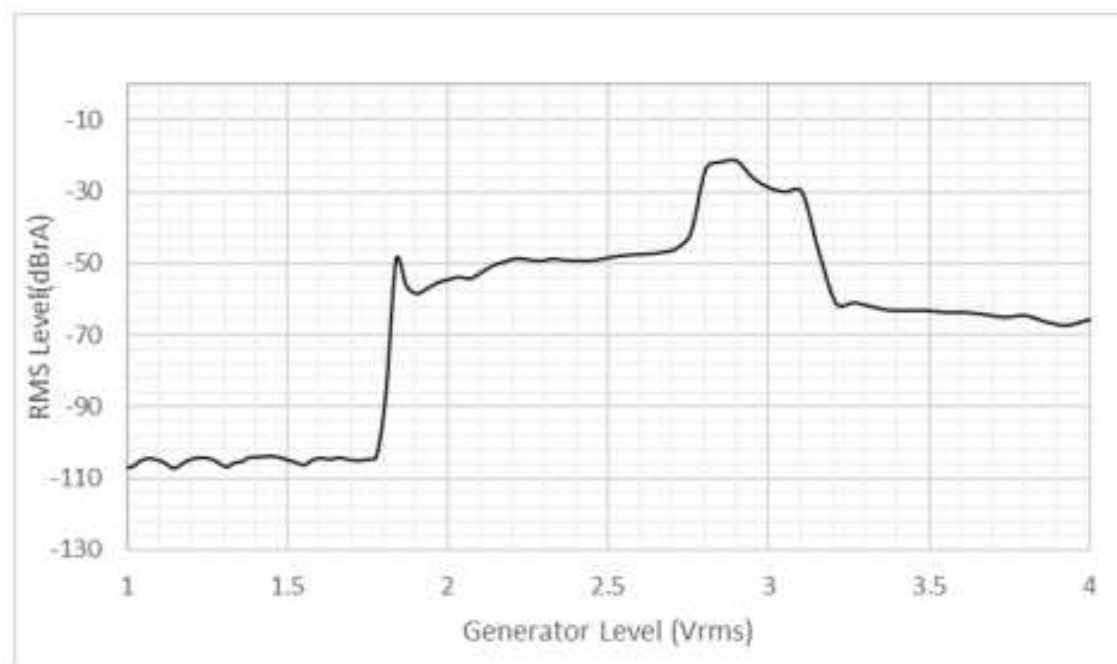


Fig 6. THD + N vs. Input with 1kΩ Load

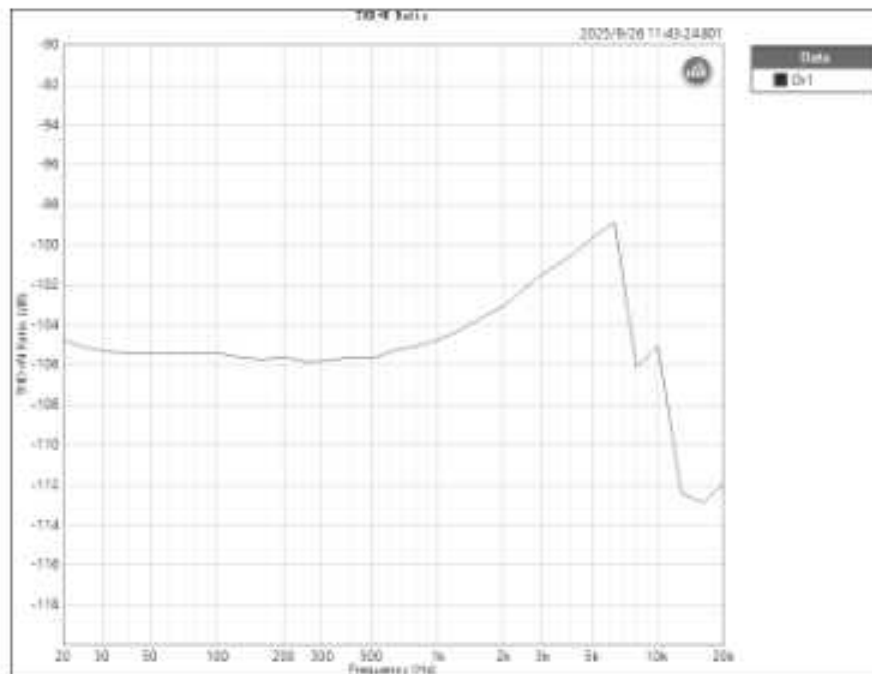


Fig 7. THD + N vs. Frequency with 33Ω Load

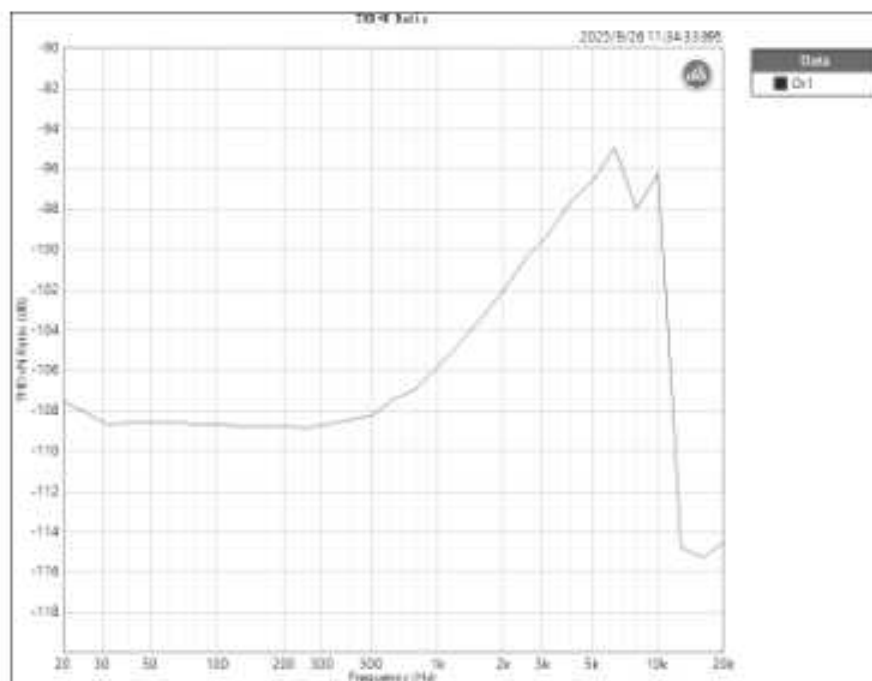


Fig 8. THD + N vs. Frequency with 1kΩ Load

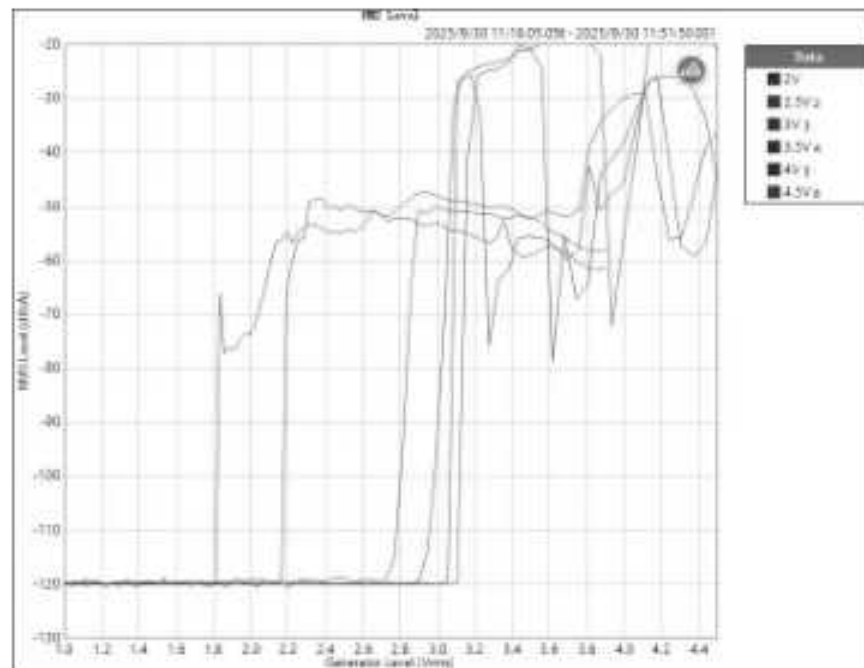


Fig 9. Off-isolation vs. Input with 32Ω Load

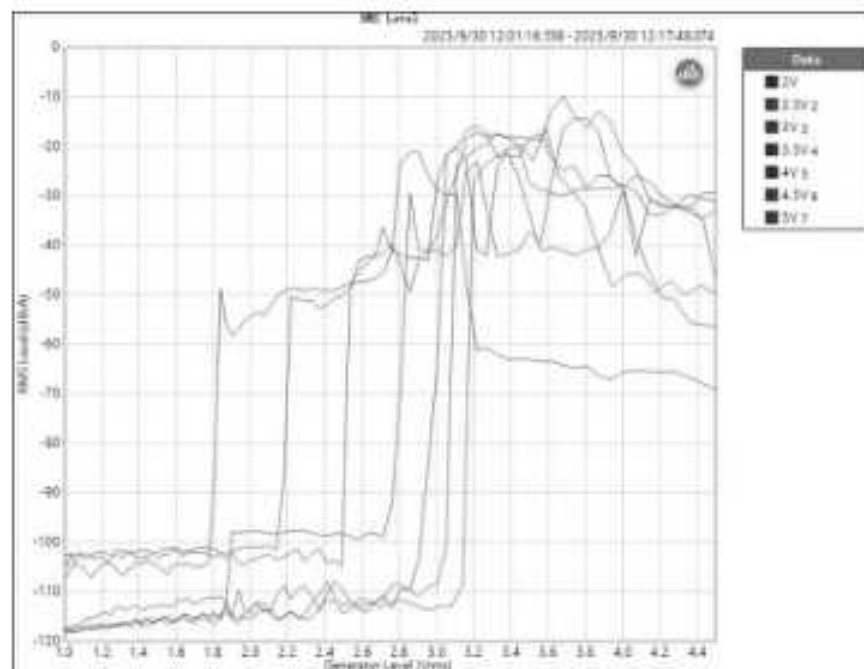
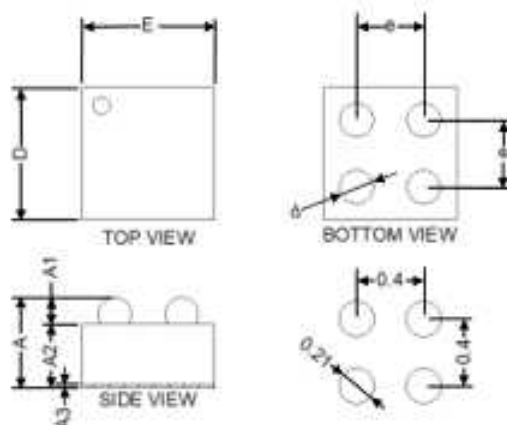


Fig 10. Off-isolation vs. Input with 1kΩ Load

## 12. Package Outline

### WLCSP4



RECOMMENDED LAND PATTERN(unit:mm)

| Symbol | Dimensions in Millimeters |       |       |
|--------|---------------------------|-------|-------|
|        | Min.                      | Typ.  | Max.  |
| A      | 0.395                     | 0.445 | 0.495 |
| A1     | 0.144                     | 0.16  | 0.176 |
| A2     | 0.255                     | 0.285 | 0.315 |
| A3     | 0.02                      | 0.025 | 0.03  |
| D      | 0.9                       | 0.93  | 0.96  |
| E      | 0.9                       | 0.93  | 0.96  |
| e      | 0.35                      | 0.4   | 0.45  |
| d      | 0.19                      | 0.21  | 0.23  |

## 13. Tape and Reel Information

### 13.1. Carrier tape dimensions

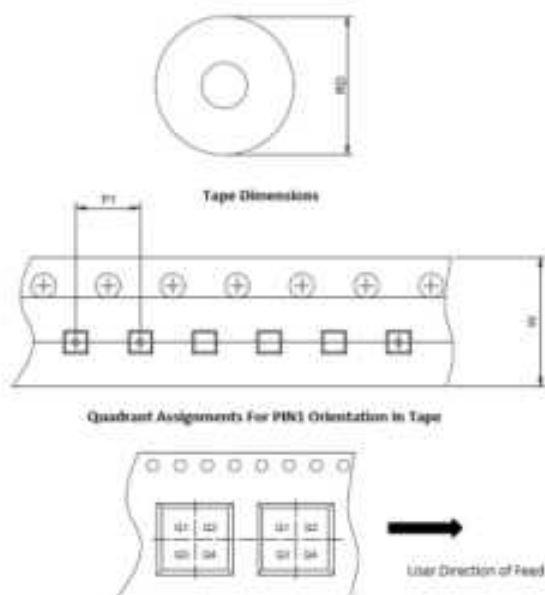
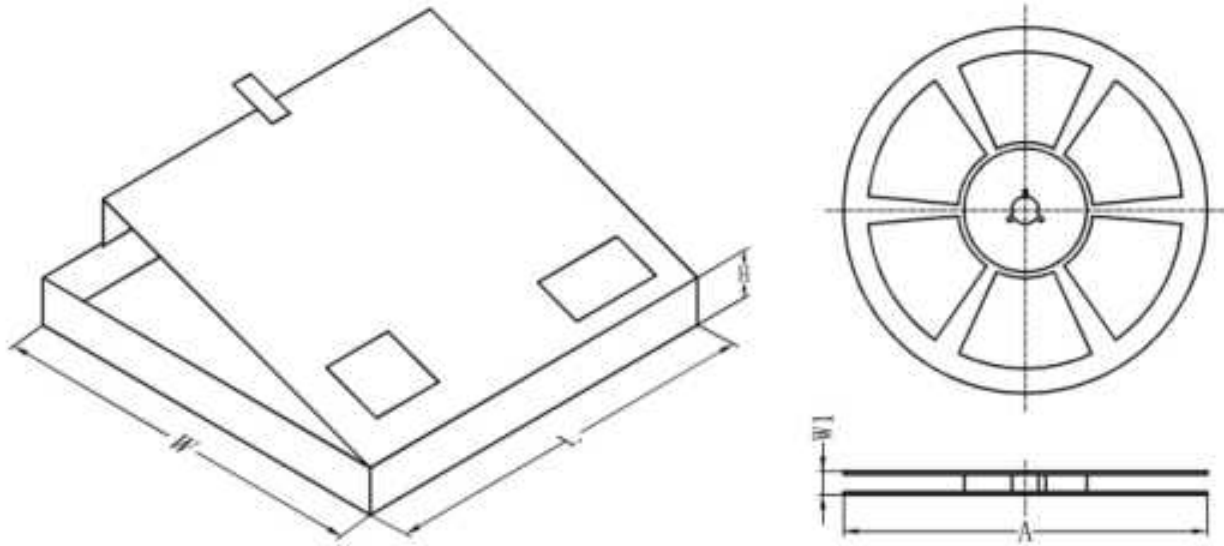


Table 7. Carrier tape dimensions

| Package version | RD(mm) | P1(mm) | W(mm) | PIN 1 |
|-----------------|--------|--------|-------|-------|
| WLCSP4          | 180    | 4      | 8     | Q1    |

## 13.2. Carrier tape dimensions



**Table 8. Dimensions and quantities**

| Package version | Type NO. ending | Reel Dimension A (mm) | Reel Width W1 (mm) | SPQ (pcs)[1] | Reels per box | Outer box dimensions L×W×H(mm)[2] |
|-----------------|-----------------|-----------------------|--------------------|--------------|---------------|-----------------------------------|
| WLCSP4          | YFQ             | 180                   | 12.2               | 3000         | 1             | 210x210x195                       |

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

## 14. Revision History

Table 9. Revision history

| Document ID      | Release Date                                                                                                                                                                                                                                  | Data sheet status | Change notice | Supersedes       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------|------------------|
| EMS3515 Rev. 1.2 | Mar 04, 2026                                                                                                                                                                                                                                  | Product datasheet |               | EMS3515 Rev. 1.1 |
| Modifications:   | <ul style="list-style-type: none"> <li>Table 4: <math>V_{DD}</math> and <math>V_{SW}</math> updated.</li> <li>Table 5: <math>V_{DD}</math> and <math>V_{SW}</math> updated.</li> <li>Section 4: Added topside marking information.</li> </ul> |                   |               |                  |
| EMS3515 Rev. 1.1 | Nov 06, 2025                                                                                                                                                                                                                                  | Product datasheet |               | EMS3515 Rev. 1.0 |
| Modifications:   | <ul style="list-style-type: none"> <li>Table 6 : Quiescent current <math>I_{DD}</math> and <math>V_{DD}</math> high voltage <math>V_{DDH}</math> updated.</li> <li>Section 11: Added figures.</li> </ul>                                      |                   |               |                  |
| EMS3515 Rev. 1.0 | Oct 21, 2025                                                                                                                                                                                                                                  | Product datasheet |               |                  |